

Text Feature Handling for Inferential Statistics

Feature Transformation → goal is to make the text feature set compatible for the model algo / NN

use-case: SMS spam classification

Given dataset

↓ (pure string)	
sms	label (categorical)
welcome to my world	s
NLP is interesting	h
NLW is awesome	h
welcome welcome	s

feature col
↓
· NUMERICAL
· CATEGORICAL

step 1: Extract the vocabulary of the corpus.

a single record in the text feature col.

DOCUMENT

→ welcome to my world
→ NLP is interesting
→ NLW is awesome
→ welcome welcome

CORPUS

set of documents in a dataset

[welcome, world, NLP, interesting, NLW, awesome]

VOCABULARY

they are unique words in a corpus without stopwords.

word
Tokenization

['welcome', 'to', 'my', 'world', 'NLP', 'is', 'interesting', 'NLW', 'is', 'awesome', 'welcome', 'welcome']

↓
Extract unique words.

['welcome', 'to', 'my', 'world', 'NLP', 'is', 'interesting', 'NLW', 'awesome']

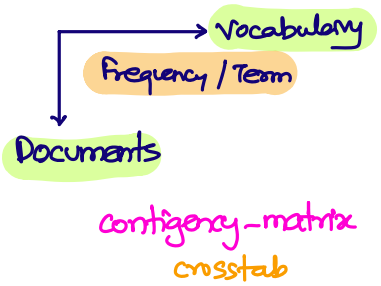
←
Remove

Stopwords

Step 2 Create BOW [Bag of words] / DTM [Document Term Matrix]

BOW is a model.

BOW represents text as a collection of words, disregarding the grammar and word order but keeping the multiplicity of the word.



VOCABULARY

document

	welcome	world	NLP	interesting	NLU	awesome
welcome to my world	1	1	0	0	0	0
NLP is interesting	0	0	1	1	0	0
NLU is awesome	0	0	0	0	1	1
welcome welcome	2	0	0	0	0	0

Frequency / Term → no. of occurrence of each vocabulary in the given document.

DOCUMENT-TERM matrix (statistics)

Step 3: Perform TF-IDF transformation [OPTIONAL]

TF → Term Frequency
It measures how frequently a term (word) appears in a document (sentence, paragraph, word).

Formula

$$TF(t, d) = \frac{\text{Number of times the term 't' appears in the document 'd'}}{\text{Total no. of terms in the document 'd'}}$$

IDF → Inverse Document Frequency

It measures how important a term is across multiple document (corpus).

Formula

$$IDF(t, D) = \log \left(\frac{\text{Total no. of documents}}{\text{Number of documents containing term 't'}} \right)$$

So TF-IDF looks,

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) * \text{IDF}(t, D)$$

$t \rightarrow$ term (word)

$d \rightarrow$ individual document (record)

$D \rightarrow$ set of relevant/all document (corpus)

eg: To calc TF

		VOCABULARY					
		welcome	world	NLP	interesting	NLU	awesome
①	→ welcome to my world	1/4	1/4	0	0	0	0
	NLP is interesting	0	0	1/3	1/3	0	0
	NLU is awesome	0	0	0	0	1/3	1/3
②	→ welcome welcome	2/2	0	0	0	0	0

$$\text{TF}(\underset{t}{\text{'welcome'}}, \underset{d}{\text{'welcome to my world'}}) = \frac{1}{4} //$$

Formula

$$\text{TF}(\underset{t}{t}, \underset{d}{d}) = \frac{\text{Number of times the term 't' appears in the document 'd'}}{\text{Total no. of terms in the document 'd'}}$$

$$\text{TF}(\underset{\uparrow}{\text{'welcome'}}, \underset{\uparrow}{\text{'welcome welcome'}}) = \frac{2}{2} = 1 //$$

To calc IDF,

Formula

$$\text{IDF}(\underset{t}{t}, \underset{D}{D}) = \log \left(\frac{\text{Total no of documents}}{\text{Number of documents containing term 't'}} \right)$$

$$\text{IDF}(\text{welcome}, \text{corpus}) = \log \left(\frac{4}{2} \right) = 0.69$$

$$\text{IDF}(\text{world}, \text{corpus}) = \log \left(\frac{4}{1} \right) = 1.39$$

TF-IDF matrix

	feature						label
	welcome	world	NLP	interesting	NLU	awesome	
→ welcome to my world	0.173	0.347	0	0	0	0	S
NLP is interesting	0	0	0.462	0.462	0	0	h
NLU is awesome	0	0	0	0	0.462	0.462	h
welcome welcome	0.693	0	0	0	0	0	S

d = NLP is interesting
 1 2 3

$$\begin{aligned}
 \text{TF}(\text{NLP}, d) * \text{IDF}(\text{NLP}, D) &= 0.333 * 1.386 \\
 &= 0.462
 \end{aligned}$$

$$\text{TF}(t, d) = \frac{\text{Number of times the term } \text{NLP} \text{ appears in the document 'd'}}{\text{Total no. of terms in the document 'd'}}$$

$$= \frac{1}{3} = 0.333$$

$$\text{IDF}(\text{NLP}, D) = \log_2\left(\frac{4}{1}\right) = 1.386$$

